Enhancing Metaheuristics for Pistachio Supply Chain Optimization

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Abstract

Efficient supply chain management is critical for high-value agricultural products. While mixed-integer linear programming models for the pistachio industry exist, the literature lacks efficient heuristic approaches with clear resolution methodologies. This paper addresses this issue by presenting an enhanced Variable Neighborhood Search (VNS) approach. Unlike prior studies that used real-valued encoding with unspecified decoding steps, we propose a transparent integer encoding scheme and a rigorous decoding algorithm. Furthermore, we develop domain-specific neighborhood structures that significantly improve search convergence compared to classical operators. We benchmark our method against Genetic Algorithms (GA), Iterated Local Search (ILS), and an exact solver. The experiments reveal that while the exact method provide optimal solutions for small instances, the proposed VNS is robust and superior in finding high-quality solutions for large-scale datasets, outperforming other metaheuristics in both stability and objective function values.

Keywords: Supply Chain Optimization, Metaheuristics, Variable Neighborhood Search, Sustainable Logistics

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1. Introduction

The growing role of agriculture in the economic development of many countries highlights the importance of integrating this sector with other industries to stimulate overall economic growth. Agriculture not only supports food security but also acts as a major contributor to exports, generating significant income and strengthening other sectors. Among various agricultural products, pistachios stand out due to their high economic and nutritional value. Major producers such as the US, Turkey, and Iran dominate global pistachio production, remaining the leading producers worldwide in 2022 and 2023 (Statista (2023)). As a result, the strategic importance of pistachios in international trade continues to grow.

With the increasing global demand for pistachios, the need for efficient supply chain networks becomes more critical. Proper supply chain management is vital for maintaining product quality, minimizing environmental impacts, and ensuring the smooth flow of goods from production to consumers. A key challenge in pistachio production is the management of its by-products, such as hulls and shells, which, while free of heavy metals or pathogens, can degrade soil quality and contribute to environmental pollution if not properly handled (Doula et al. (2014); Bartzas and Komnitsas (2017)). Furthermore, improper disposal of these materials may lead to the growth of molds that increase the risk of aflatoxin contamination, posing serious health risks such as liver cancer (Hokmabadi (2018); Dronavalli and Kang (2019)). Therefore, eco-friendly practices for reusing agricultural waste are essential to mitigate both environmental and health hazards.

Moreover, inefficiencies in supply chain networks for food and agricultural products, including pistachios, can have significant implications for product quality and trade (Casper and Sundin (2018)). Addressing these environmental and logistical concerns is crucial to sustaining both the economic and ecological viability of the pistachio industry. In the broader context of sustainable supply chains, several studies have focused on optimizing resource use and minimizing waste through recovery and recycling practices. For example, sustainable supply chain models designed for agriculture and other industries have shown significant environmental benefits by reducing waste and reusing by-products for processes like biogas and compost production (Cheraghalipour and Roghanian (2022)). Such approaches have been successfully applied in sectors such as manufacturing and waste recovery, where efficient resource management and cost reduction have been key drivers (Devika et al. (2014); Zohal and Soleimani (2016)). These strategies underline the importance of sustainability in modern supply chains and the potential for agricultural industries to implement similar measures.

In related research, Rajabi-Kafshgar et al. (2023) focused on optimizing the pistachio supply chain using various algorithms, primarily population-based approaches such as the Genetic Algorithm (GA) (Fathollahi-Fard and Hajiaghaei-Keshteli (2018); Xie et al. (2022)), among others. However, a significant limitation in their work is the lack of methodological transparency. Specifically, the authors employed a real-valued encoding scheme without detailing the decoding procedure required to transform these values into a feasible transport tree. This omission hinders reproducibility and obscures the mechanism by which constraints are handled. Furthermore, the literature has increasingly questioned the effectiveness of "metaphor-based" metaheuristics (such as the Imperialist Competitive Algorithm and the Social Engineering Optimizer, used by the authors), suggesting that established methods often perform equally well or better without the complexity of novel analogies (Campelo and Aranha (2023); Aranha et al. (2022)).

Addressing these gaps, this research focuses on designing an optimal supply chain network for pistachios by leveraging the Variable Neighborhood Search (VNS) algorithm (Gendreau and Potvin (2010)). While we adopt the same mixed-integer linear programming (MILP) model formulation from Rajabi-Kafshgar et al. (2023), we propose a robust overhaul of the resolution methodology. We introduce a transparent integer-based priority encoding and provide a step-by-step description of the decoding algorithm, ensuring reproducibility. Furthermore, we develop problem-specific neighborhood operators designed to exploit the particular structure of the chain. These specialized operators are contrasted with classical moves to demonstrate their efficacy in exploring the solution space.

To validate the proposed approach, we conduct a comprehensive comparative analysis against a Genetic Algorithm (GA), an Iterated Local Search (ILS), and an exact Branch-and-Cut method (using the Gurobi solver). Our findings demonstrate that the proposed VNS with specialized operators consistently outperforms the other metaheuristics in terms of solution quality. On top of that, we show that for large-scale instances, where exact methods become infeasible due to memory limitations and excessive runtime, the proposed heuristic offers a vital trade-off, providing high-quality (and sometimes, even better) feasible solutions efficiently.

The remainder of this paper is organized as follows. Section 2 presents the problem definition and the mathematical formulation. Section 3 details the heuristic approach, introducing the integer encoding, the specific decoding steps, and the novel domain-specific neighborhood structures. Section 4 reports the computational experiments, offering a statistical comparison between VNS, GA, ILS, and the exact solver across varying instance sizes. Finally, Section 5 summarizes the key findings and suggests future research directions.

2. Problem Definition

The logistics network proposed by Rajabi-Kafshgar et al. (2023), illustrated in Figure 1, describes the flow of products and by-products along the pistachio supply chain. The process begins with the farmers (i), who send the harvested pistachios to the processing centers (j). At these centers, the pistachios are transformed into three main components: open-shell pistachios, raw kernels, and organic waste.

Each of these components follows specific flows. Open-shell pistachios are forwarded to pistachio factories (k), which process and distribute them to final customers (n_1). Raw kernels are sent to oil extraction centers (e), responsible for producing pistachio oil, which is directed to both final customers (n_2) and cosmetic factories (s). These, in turn, process the oil into cosmetic products that are supplied to their respective consumers (n_3). Organic waste generated in the initial processing is sent to composting centers (q), where it is transformed into compost, subsequently distributed to composting customers (m).

Although, in practical terms, the produced compost could be reused by the original farmers, establishing a closed-loop system, the adopted mathematical model does not explicitly represent this feedback. Instead, compost customers are not directly associated with the inital producers, characterizing an implicit feedback cycle. This approach, already adopted in previous works (Pishvaee et al. (2010); Ahmadi and Amin (2019); Banasik et al. (2016)), preserves the structural simplicity of the model. However, the inclusion of rigorous modeling of this return loop represents a promising direction for future investigations, especially in the context of sustainable supply chains.

The mixed linear problem aims to minimize total costs (1), including opening, production, and transportation costs, subject to the following capacity and demand constraints, tolerating no shortages (see more details on the variables in Table 5 of Rajabi-Kafshgar et al. (2023)):

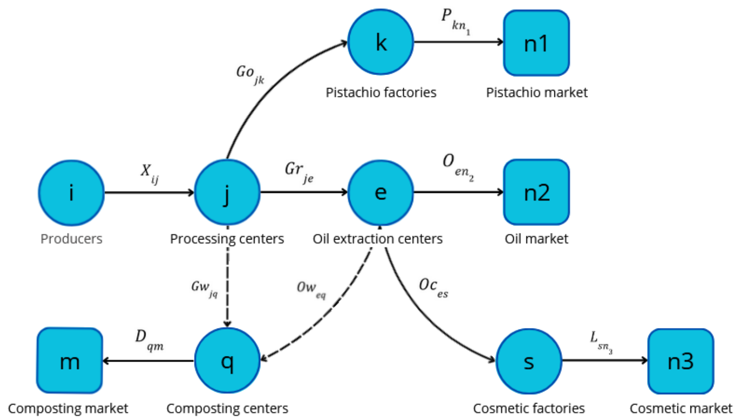

Figure 1: Structure of the proposed pistachio supply chain.

$$\min \; z_1 + z_2 + z_3 \tag{1}$$

$$\text{s.t.:} \quad \sum_{j=1}^{J} X_{ij} \leq Cpa_i, \quad \forall i \in I \tag{2}$$

$$\sum_{i=1}^{I} X_{ij} \leq Cpu_j \times U_j, \quad \forall j \in J \tag{3}$$

$$\sum_{e=1}^{E} Ow_{eq} + \sum_{j=1}^{J} Gw_{jq} \leq Cpy_q \times Y_q, \quad \forall q \in Q \tag{4}$$

$$\sum_{j=1}^{J} Go_{jk} \leq Cpw_k \times W_k, \quad \forall k \in K \tag{5}$$

$$\sum_{j=1}^{J} Gr_{je} \leq Cpr_e \times R_e, \quad \forall e \in E \tag{6}$$

$$\sum_{e=1}^{E} O_{en_2} \leq Cpv_s \times V_s, \quad \forall s \in S \tag{7}$$

$$\sum_{k=1}^{K} Go_{jk} + \sum_{e=1} Gr_{je} + \sum_{q=1}^{Q} Gw_{jq} \leq \sum_{i=1} X_{ij}, \quad \forall j \in J \tag{8}$$

$$\sum_{k=1}^{K} Go_{jk} = (1-\beta) \sum_{i=1}^{I} X_{ij}\theta_1, \quad \forall j \in J \tag{9}$$

$$\sum_{e=1}^{K} Gr_{je} = (1-\beta) \sum_{i=1}^{I} X_{ij}\theta_2, \quad \forall j \in J \tag{10}$$

$$\sum_{q=1}^{Q} Gw_{jq} = \sum_{i=1}^{I} X_{ij}\theta_3, \quad \forall j \in J \tag{11}$$

$$\theta_1 + \theta_2 + \theta_3 = 1 \tag{12}$$

$$\sum_{n_1=1}^{N_1} P_{kn_1} \leq \sum_{j=1}^{J} Go_{jk} \times \gamma k, \quad \forall k \in K \tag{13}$$

$$\sum_{n_2=1}^{N_2} O_{en_2} + \sum_{s=1}^{S} Oc_{es} \leq \sum_{j=1} Gr_{je} \times (1-\lambda), \quad \forall e \in E \tag{14}$$

$$\sum_{q=1}^{Q} Ow_{eq} \leq \sum_{j=1}^{J} Gr_{je} \times \lambda, \quad \forall e \in E \tag{15}$$

$$\sum_{n_3=1}^{N_3} L_{sn_1} \leq \sum_{e=1}^{E} Oc_{es} \times \gamma s, \quad \forall s \in S \tag{16}$$

$$\sum_{m=1}^{M} D_{qm} \leq \left(\sum_{j=1}^{J} Gw_{jq} + \sum_{e=1}^{E} Ow_{eq} \right) \times \gamma q, \quad \forall q \in Q \tag{17}$$

$$\sum_{k=1}^{K} P_{kn_1} \geq Dp_{n_1}, \quad \forall n_1 \in N_1 \tag{18}$$

$$\sum_{e=1}^{E} O_{en_2} \geq Du_{n_2}, \quad \forall n_2 \in N_2 \tag{19}$$

$$\sum_{s=1}^{S} L_{sn_3} \geq Ds_{n_3}, \quad \forall n_3 \in N_3 \tag{20}$$

$$\sum_{q=1}^{Q} D_{qm} \leq Dc_m, \quad \forall m \in M \tag{21}$$

- The objective function (1) minimizes the total costs of the supply chain, including opening, transportation, and production costs. The costs are divided into three components: z_1 (22), z_2 (23), and z_3 (24), which represent facility opening costs, transportation costs, and production costs, respectively, as shown in the following equations:

$$
\begin{aligned}
z_1 = {} & \sum_{j=1}^{J} Fu_j \cdot U_j + \sum_{q=1}^{Q} Fy_q \cdot Y_q + \sum_{k=1}^{K} Fw_k \cdot W_k \\
& + \sum_{e=1}^{E} Fr_e \cdot R_e + \sum_{s=1}^{S} Fv_s \cdot V_s
\end{aligned} \tag{22}
$$

$$
\begin{aligned}
z_2 = {} & \sum_{i=1}^{I} \sum_{j=1}^{J} CI_i \cdot X_{ij} + \sum_{j=1}^{J} \sum_{k=1}^{K} Cu_{1j} \cdot Go_{jk} \\
& + \sum_{j=1}^{J} \sum_{e=1}^{E} Cu_{2j} \cdot Gr_{je} + \sum_{q=1}^{Q} \sum_{m=1}^{M} Cy_q \cdot D_{qm} \\
& + \sum_{k=1}^{K} \sum_{n_1=1}^{N_1} Cw_k \cdot P_{kn_1} \\
& + \sum_{e=1}^{E} Cr_e \cdot \left(\sum_{n_2=1}^{N_2} O_{en_2} + \sum_{s=1}^{S} Oc_{es} \right) \\
& + \sum_{s=1}^{S} \sum_{n_3=1}^{N_3} Cv_s \cdot L_{sn_3}
\end{aligned} \tag{23}
$$

$$
\begin{aligned}
z_3 = {} & \sum_{i=1}^{I} \sum_{j=1}^{J} CX_{ij} \cdot X_{ij} + \sum_{j=1}^{J} \sum_{k=1}^{K} CK_{jk} \cdot Go_{jk} \\
& + \sum_{j=1}^{J} \sum_{q=1}^{Q} CJ_{jq} \cdot Gw_{jq} + \sum_{e=1}^{E} \sum_{s=1}^{S} CS_{es} \cdot Oc_{es} \\
& + \sum_{e=1}^{E} \sum_{n_2=1}^{N_2} CN_{en_2} \cdot Oc_{en_2} + \sum_{e=1}^{E} \sum_{q=1}^{Q} CQ_{eq} \cdot Ow_{eq} \\
& + \sum_{s=1}^{S} \sum_{n_3=1}^{N_3} Cl_{sn_3} \cdot L_{sn_3} + \sum_{k=1}^{K} \sum_{n_1=1}^{N_1} Cp_{kn_1} \cdot P_{kn_1} \\
& + \sum_{q=1}^{Q} \sum_{m=1}^{M} Cd_{qm} \cdot D_{qm}
\end{aligned} \tag{24}
$$

- The variables R_j, Y_q, W_k, Z_e, and V_s are binary and determine whether certain elements of the model are active. For example, $R_j = 1$ indicates that facility j is operational, and $R_j = 0$ otherwise. The other variables in the model are continuous and non-negative, representing real values greater than or equal to zero. These variables model flows, quantities, and other measurable factors within the system.

- Constraints (2)-(7) address the capacity and demand limitations of each facility in the supply chain. In particular, constraint (2) limits the quantity of products transported from farmers to distribution centers. Constraints (3)-(7) ensure that the quantities sent to processing centers, composting centers, pistachio factories, oil extraction centers, and cosmetic factories do not exceed the capacities of these facilities.

- The quantity of products shipped from any facility must not exceed the orders received by that facility. Consequently, constraint (8) establishes that the quantity of products shipped from a processing center must be less than or equal to the quantity received by other facilities. Constraints (9)-(12) define the quantities of open-shell pistachios, raw kernels, and waste generated during the hulling process at each processing center. Similarly, constraints (13)-(17) govern the flows within pistachio factories, oil extraction centers, cosmetic factories, and composting centers.

- Finally, to ensure that customer demand is met, constraints (18)-(21) ensure that the quantities of products delivered to pistachio, pistachio oil, cosmetics, and compost customers satisfy or exceed their respective demands.

3. Methodology

This section outlines the heuristic methodology employed to solve the optimization problem. The proposed approach integrates a Variable Neighborhood Search (VNS) algorithm with a robust priority-based encoding scheme. Our method introduces a transparent integer-based representation and a backward decoding strategy to ensure feasibility across the whole chain. Furthermore, to enhance the search capability, we develop and apply problem-specific neighborhood structures designed to exploit the specific constraints of the pistachio supply chain.

3.1. Solution Representation

Representing a solution for a complex network design problem is nontrivial. Following the principles established by Gen et al. (2006) and Rajabi-Kafshgar et al. (2023), we employ a priority-based encoding method. In this scheme, the chromosome does not directly encode the flow quantities (X_{ij}), but rather the *priority* of serving specific transportation arcs.

3.1.1. Integer Permutation vs. Real-Valued Encoding

In this work, it is used an integer permutation encoding rather than the random key ($0-1$ real value) encoding used in previous studies (Rajabi-Kafshgar et al. (2023); Fathollahi-Fard and Hajiaghaei-Keshteli (2018)).

In the real-valued approach, genes are assigned random numbers $r \in [0, 1]$, and priorities are derived from sorting these values. This representation suffers from a "many-to-one" mapping redundancy: infinite combinations of real numbers can result in the exact same priority order. For example, the vectors $v_1 = [0.1, 0.2, 0.5]$ and $v_2 = [0.8, 0.85, 0.9]$ both map to the same permutation indices $(1, 2, 3)$. This redundancy creates "flat" regions in the search landscape where small perturbations (mutations) in the genotype produce no change in the phenotype (the solution structure).

In contrast, we employ sequential integers from 1 to Z, where Z is the total number of candidate source-depot pairs in a specific chain segment. This ordinal representation ensures a one-to-one mapping between the genotype and the solution structure. This property is crucial for the VNS algorithm, as it ensures that every neighborhood move results in a distinct solution configuration, effectively eliminating redundant search steps.

3.1.2. Chromosome Structure

The solution is represented by a composite chromosome divided into eight distinct segments (S_1 to S_8), as illustrated in Fig. 2. Each segment contains a permutation of integers representing the transport priorities for a specific part of the supply chain.

3.2. Decoding Procedure

The decoding process acts as a heuristic builder that transforms the priority chromosome into a feasible transportation tree. As shown in Fig. 3, the decoding process begins with the flow of final products (pistachios, oil, cosmetics) from factories to consumers. This ensures that all constraints are

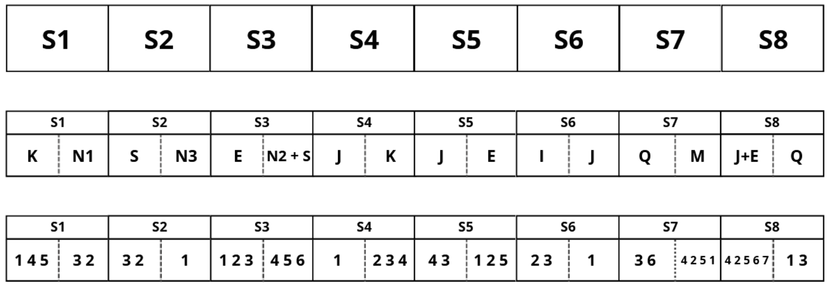

Figure 2: Structure of the priority-based chromosome. The solution is composed of 8 distinct integer permutations ($S_1 - S_8$), each governing the transport priority of a specific subsection in the supply chain.

met: consumer demand is satisfied first, propagating the requirements upstream to factories and processing centers, ensuring that facility capacities are respected at every stage.

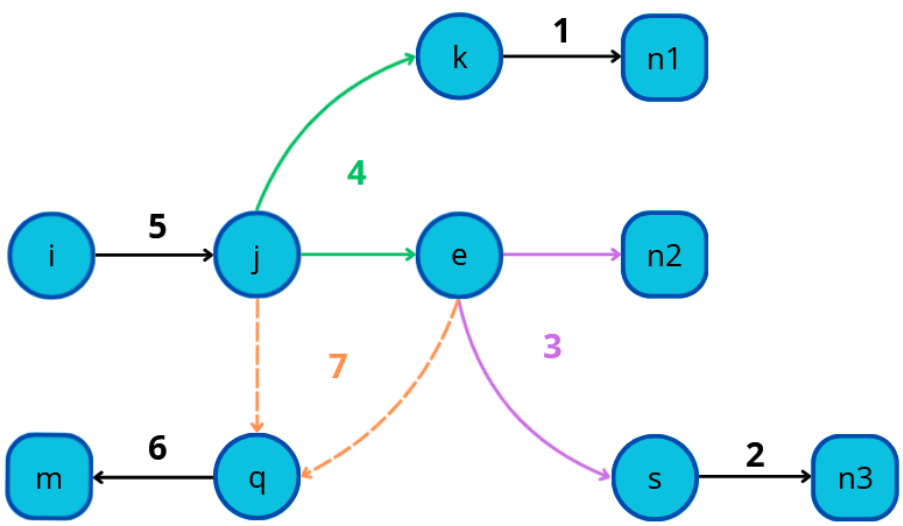

Figure 3: The backward decoding flow. The process prioritizes satisfying end-consumer demands first to propagate requirements upstream, ensuring feasibility throughout the chain.

Algorithm 1 details the first phase of this process. The procedure utilizes a helper function, *decodingStep*, which implements the classic priority-based transportation logic: it allocates flow from sources to destinations based on the priority order defined in the chromosome segment (S_k), always selecting the lowest-cost available route.

The critical logic occurs in Steps 5 and 6 (lines 14-17). Here, the algorithm must reconcile the diverging flows from Processing Centers. Since Processing Centers produce multiple outputs (Kernels for Oil vs. Open-shell for Factories) in fixed proportions, satisfying the demand of one downstream entity often creates a surplus of the other. Our decoding logic identifies the bottleneck demand and routes any inevitable surplus to the destination with the lowest transportation cost, minimizing the penalty for over-supply.

Once the intermediate flows are established, the algorithm proceeds to the upstream and waste management stages, as detailed in Algorithm 2.

First, the exact demand for raw pistachios at the Processing Centers is calculated based on the outputs determined in Phase 1 (lines 2-5). Segment S_6 is then used to fulfill this demand from Farmers. If the aggregate capacity of farmers is insufficient, the solution is marked as infeasible.

Finally, waste management is handled in Steps 9 and 10. The waste generated at Processing and Oil Extraction centers is "pushed" to Composting Centers. Segment S_8 determines the priority of these flows, directing waste to the Composting Center with available capacity and the lowest associated cost.

3.3. The VNS Algorithm

The proposed algorithm is a hybrid metaheuristic that enhances the standard Variable Neighborhood Search (VNS) (Hansen et al. (2010)) by integrating Variable Neighborhood Descent (VND) for intensification and a Simulated Annealing (SA) acceptance criterion. This hybridization addresses the limitations of pure descent methods in the highly constrained landscape of the pistachio supply chain. The procedure is outlined in Algorithm 3.

The intensification phase employs a Variable Neighborhood Descent (VND) strategy rather than a simple local search. In this step, the algorithm iterates sequentially through the entire set of neighborhood structures. Whenever an improvement is found with any operator, the search restarts from the first operator, ensuring that the resulting solution is a local optimum with respect to all defined neighborhoods.

To prevent stagnation in local optima, we replace the standard deterministic acceptance criterion with a probabilistic Simulated Annealing mechanism. A new solution generated by the VND phase is accepted as the current

Algorithm 1 Chromosome decoding for the supply chain (Part 1)

procedure DECODE(data)
  Input: structured data containing:
  - Capacities (C) and demands (D) for all facilities
  - Unit transportation costs (T)
  - Loss rates for each process stage
  - Consumer demands
  - Chromosome segments ($S_1, S_2, \ldots, S_8$) representing transport priorities
  Output: TotalCost, Decision Variables ($X, P, L, O, \ldots$)
  // **Step 1: Initialize variables**
  $TotalCost \leftarrow 0$
  // **Step 2: Decode Pistachio Factories $\rightarrow$ Pistachio Consumers**
  $a_1 \leftarrow$ capacities of pistachio factories
  $b_1 \leftarrow$ demands of pistachio consumers
  $c_1 \leftarrow$ transportation costs
  $(cost1, P) \leftarrow$ decodingStep(S_1, a_1, b_1, c_1)
  $TotalCost \leftarrow TotalCost + cost1$
  // **Step 3: Decode Cosmetic Factories $\rightarrow$ Cosmetic Consumers**
  $a_2 \leftarrow$ capacities of cosmetic factories
  $b_2 \leftarrow$ demands of cosmetic consumers
  $c_2 \leftarrow$ transportation costs
  $(cost2, L) \leftarrow$ decodingStep(S_2, a_2, b_2, c_2)
  $TotalCost \leftarrow TotalCost + cost2$
  // **Step 4: Decode Oil Extraction Centers $\rightarrow$ Consumers**
  $a_3 \leftarrow$ capacities of oil extraction centers
  $b_3 \leftarrow$ demands of oil consumers + cosmetic factories
  $c_3 \leftarrow$ transportation costs
  $(cost3, OO_c) \leftarrow$ decodingStep(S_3, a_3, b_3, c_3)
  Separate OO_c into O (oil consumers) and O_c (cosmetic factories)
  $TotalCost \leftarrow TotalCost + cost3$
  // **Step 5: Decode Processing Center $\rightarrow$ Pistachio Factories**
  $a_4 \leftarrow$ capacities of processing centers
  $b_4 \leftarrow$ adjusted demands of pistachio factories
  $c_4 \leftarrow$ transportation costs
  $(\_, G_o) \leftarrow$ decodingStep(S_4, a_4, b_4, c_4)
  // **Step 6: Decode Processing Center $\rightarrow$ Oil Extraction Centers**
  $a_5 \leftarrow$ capacities of processing centers
  $b_5 \leftarrow$ adjusted demands of oil extraction centers
  $c_5 \leftarrow$ transportation costs
  $(\_, G_r) \leftarrow$ decodingStep(S_5, a_5, b_5, c_5)
end procedure

Algorithm 2 Chromosome decoding for the supply chain (Part 2)

procedure DECODE(continuation)
  // Step 7: Adjust demands at Processing Centers
  for each processing center j **do**
    Calculate required outputs based on pistachio and oil flows
    Update G_o and G_r to match effective capacities
  end for
  // Step 8: Decode Pistachio Farmers → Processing Centers
  $a_6 \leftarrow$ capacities of pistachio farmers
  $b_6 \leftarrow$ updated demands at processing centers
  $c_6 \leftarrow$ transportation costs
  $(cost6, X) \leftarrow$ decodingStep(S_6, a_6, b_6, c_6)
  $TotalCost \leftarrow TotalCost + cost6$
  // Step 9: Decode Composting Centers → Compost Consumers
  $a_7 \leftarrow$ capacities of composting centers
  $b_7 \leftarrow$ demands of compost consumers
  $c_7 \leftarrow$ transportation costs
  $(cost7, D) \leftarrow$ decodingStep(S_7, a_7, b_7, c_7)
  $TotalCost \leftarrow TotalCost + cost7$
  // Step 10: Assign waste flows
  Calculate waste from processing centers (J) and Oil Centers (E)
  $a_8 \leftarrow$ available capacities of composting centers
  $b_8 \leftarrow$ waste demands to be disposed
  $c_8 \leftarrow$ transportation costs
  $(\_, O_w) \leftarrow$ decodingStep(S_8, a_8, b_8, c_8)
  Update $TotalCost$
  // Step 11: Define binary opening variables
  Set $U, W, R, V, Y = 1$ if corresponding flow > 0
  Add fixed costs to $TotalCost$
  return $TotalCost$, Decision Variables, Binary Variables
end procedure

solution if it improves the objective function or, if it is worse, with a probability given by the Boltzmann factor $P = e^{-\Delta/T}$, where Δ is the deterioration in solution quality and T is the current temperature. This allows the search to traverse valleys in the fitness landscape.

Finally, the shaking phase applies a perturbation to the current solution to escape the basin of attraction of the local optimum. The magnitude of this perturbation is adaptive, defined in our implementation as 7% of the total decision variables in the chromosome. This ensures that the diversification strength scales appropriately with the problem size. The specific operator used for shaking changes cyclically, reseting to the first operator upon a successful acceptance and moves to the next operator if the candidate solution is rejected.

3.4. Neighborhood Structures

A critical inefficiency in applying standard permutation operators to priority-based encoding is their "blindness" to the phenotypic state. In the greedy decoding process (Algorithm 1), typically only a subset of high-priority nodes is required to satisfy all demands. We term these **Active Nodes** ($\mathcal{A}$), while the remaining unused nodes are **Inactive** ($\mathcal{I}$). Standard operators often swap two Inactive nodes, changing the genotype but resulting in the exact same transportation tree and objective value.

To overcome this redundancy, our implementation explicitly tracks the set of active nodes returned by the decoder and employs four structure-aware operators. These operators are designed to either intensify the search within the current topology or diversify it by forcing interactions between the active and inactive sets:

- **Active Reversion:** This operator functions as an intensification mechanism. It selects a segment of the chromosome containing multiple nodes from set $\mathcal{A}$ and reverses their priority values. By reordering only the suppliers that are already in use, this move fine-tunes the flow distribution to find lower-cost allocations without necessarily opening new routes or altering the network structure.

- **Inactive-Active Swap:** This is a global diversification operator. It selects one active node ($i \in \mathcal{A}$) and one inactive node ($j \in \mathcal{I}$) at random and exchanges their priorities. This forces a currently used route to be downgraded and a previously unused route to be upgraded,

Algorithm 3 Hybrid VNS with SA Acceptance

Require: Initial solution x, Operators set $\mathcal{N}$, Max evaluations N_{max}, Initial Temp T_0, Cooling rate α

Output: Best solution found x^*
$x_{curr} \leftarrow x; \quad x^* \leftarrow x$
$n_{eval} \leftarrow 1; \quad T \leftarrow T_0$
$k \leftarrow 0$ ▷ Operator index for shaking
while $n_{eval} < N_{max}$ **do**
  // Phase 1: Intensification (VND)
  $x_{new} \leftarrow$ VND_LocalSearch$(x_{curr}, \mathcal{N})$
  Update n_{eval}
  // Phase 2: Update Global Best
  if $f(x_{new}) < f(x^*)$ **then**
    $x^* \leftarrow x_{new}$
  end if
  // Phase 3: Acceptance Criteria (SA)
  $\Delta \leftarrow f(x_{new}) - f(x_{curr})$
  if $\Delta \leq 0$ **or** Random$(0,1) < e^{-\Delta/T}$ **then**
    $x_{curr} \leftarrow x_{new}$
    $k \leftarrow 0$ ▷ Reset shaking operator on acceptance
  else
    $k \leftarrow (k+1) \pmod{|\mathcal{N}|}$ ▷ Next shaking operator
  end if
  // Phase 4: Perturbation and Cooling
  $T \leftarrow T \times \alpha$
  $x_{curr} \leftarrow$ Shake$(x_{curr}, \mathcal{N}[k])$
end while

effectively testing global topology changes by bringing new nodes into the solution.

- **Inactive Boost:** This novel problem-specific operator addresses the "starvation" issue common in capacity-constrained problems. A low-cost supplier might be ignored simply because it sits too low in the priority list to be considered before demands are met. This operator selects a random inactive node and assigns it the highest possible priority value in the chromosome. This forces the greedy decoder to maximize flow from this specific node before considering any others, potentially revealing a completely different and superior tree structure.

- **Inactive-Active Exchange two Neighbors (IA-ETN):** This operator provides a "localized" diversification. Instead of a random global swap, it selects an inactive node and swaps it with its nearest active neighbor in terms of chromosome index distance. Unlike the global swap, which can be highly disruptive, this operator explores the "boundary" of the solution, testing if a route that is "almost" good enough (close to the active set) should replace a marginally active route.

4. Computational Experiments

Morbi luctus, wisi viverra faucibus pretium, nibh est placerat odio, nec commodo wisi enim eget quam. Quisque libero justo, consectetuer a, feugiat vitae, porttitor eu, libero. Suspendisse sed mauris vitae elit sollicitudin malesuada. Maecenas ultricies eros sit amet ante. Ut venenatis velit. Maecenas sed mi eget dui varius euismod. Phasellus aliquet volutpat odio. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Pellentesque sit amet pede ac sem eleifend consectetuer. Nullam elementum, urna vel imperdiet sodales, elit ipsum pharetra ligula, ac pretium ante justo a nulla. Curabitur tristique arcu eu metus. Vestibulum lectus. Proin mauris. Proin eu nunc eu urna hendrerit faucibus. Aliquam auctor, pede consequat laoreet varius, eros tellus scelerisque quam, pellentesque hendrerit ipsum dolor sed augue. Nulla nec lacus.

5. Conclusion

Suspendisse vitae elit. Aliquam arcu neque, ornare in, ullamcorper quis, commodo eu, libero. Fusce sagittis erat at erat tristique mollis. Maecenas sapien libero, molestie et, lobortis in, sodales eget, dui. Morbi ultrices rutrum lorem. Nam elementum ullamcorper leo. Morbi dui. Aliquam sagittis. Nunc placerat. Pellentesque tristique sodales est. Maecenas imperdiet lacinia velit. Cras non urna. Morbi eros pede, suscipit ac, varius vel, egestas non, eros. Praesent malesuada, diam id pretium elementum, eros sem dictum tortor, vel consectetuer odio sem sed wisi.

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